

Claims and limitations

Repository source: certificate/CLAIMS_AND_LIMITATIONS.md

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Main claim represented by the package

The exact certificate and manuscript represent the universal statement

$$\text{Vol}(K) \geq \frac{26220940089713\pi}{20000000000000} d^3 = 0.41187756378030227424817892598 \dots d^3$$

for every convex body $K \subset \mathbb{R}^3$ of constant width $d > 0$.

The claim is universal: it is not restricted by symmetry, smoothness, contact type, prescribed circumradius, or proximity to a Meissner body. Scaling from width one supplies the factor d^3 .

Current verification classification

Proposed exact universal certificate - standalone internal replay.

This means:

- the finite certificate is represented by exact integer and rational objects;
- the standalone exact verifier accepts those objects;
- the assembled theorem certificate is reconstructed byte for byte;
- independent binary64 Python and base-R audits agree;
- ten reconstruction/regression tests pass;
- seven targeted corruptions are rejected;
- paper-facing constants and the lower-row table are generated or reconciled against the certificate JSON; and
- the package is still an AI-assisted, non-peer-reviewed research draft.

The word **exact** refers to the arithmetic of the finite certificate. It does not mean that the full proof has been formalized in a proof assistant or independently reviewed.

Claims established inside the finite package

Subject to the mathematical reductions in the manuscript, the exact package checks:

- gap-free circumradius coverage from $1/2$ through $\sqrt{6}/4$;
- the exact split $R_* = 305695601/500000000$;
- the unique lower-branch bottleneck 03B02;
- four exact replacement cells for the old 03A1 interval;

- reconstruction of a 98,306-vertex near-Jung fan;
- all exact fan-margin and inherited-multiplier inequalities;
- the determinant transfer and forced-core lower bound;
- the dominant-edge coordinate-width estimate on all three axes;
- three optimized antipodal cap-pair contributions;
- exact six-cap disjointness;
- a strict near-Jung handoff surplus; and
- the exact theorem coefficient and decimal enclosure.

What is not claimed

The package does **not** establish any of the following:

- that the bound is sharp;
- an equality case;
- rigidity or stability near equality;
- the identity or uniqueness of a minimizing body;
- that either Meissner body minimizes volume;
- the Meissner conjecture;
- a classification of all extremal contact configurations;
- a proof in Lean, Coq, Isabelle, or another proof assistant;
- external independent reproduction;
- peer review, journal acceptance, or publication;
- a systematic priority or novelty guarantee; or
- correctness of every statement merely because it was generated or checked by an AI system.

The explicit Meissner value

$$\frac{\pi}{12} \left(8 - 3\sqrt{3} \arccos \frac{1}{3} \right) d^3 = 0.419860045965080 \dots d^3$$

is used only as an explicit-body comparison value. It is not asserted here as the universal minimum.

Relationship to earlier bounds

The certificate coefficient exceeds, by direct arithmetic comparison, the public HYRA coefficient

$$\frac{130838246407123\pi}{10^{15}} > 0.411040473721188.$$

It also exceeds Nishioka’s $4\pi/33$ coefficient and Chakerian’s historical coefficient. These comparisons do not by themselves establish priority or publication status. The dated source review is recorded in [../proof/SOURCE_LEDGER.md](#).

Smoothness and approximation

The final theorem is stated for all convex bodies of constant width, including nonsmooth bodies. The analytic divergence step is first justified on a regular class and then passed to arbitrary bodies by the regularization and continuity argument stated in the manuscript. The finite verifier assumes that reduction; it does not independently formalize geometric measure theory or Hausdorff continuity.

Status words that may be used

Before public release and external review, acceptable descriptions include:

- proposed exact universal certificate;
- computer-assisted research draft;
- exact-rational finite verification;
- independently audited in Python and base R; and
- seeking independent mathematical verification.

Descriptions that should not be used at this stage include:

- peer reviewed;
- independently verified;
- formally verified;
- sharp or optimal;
- proof of the Meissner conjecture;
- identification of the minimizer; and
- final accepted theorem.

Corrections policy

A mathematical or certificate correction must be visible. It should receive a new commit and, after release, a new version or release tag. The old tagged artifact should remain available with a clear correction notice rather than being silently rewritten.

A change to the exact theorem coefficient, branch partition, proof objects, or verifier semantics requires:

1. a new certificate schema or documented schema revision;
2. regenerated exact and audit transcripts;
3. renewed adversarial tests;
4. paper and documentation reconciliation;
5. fresh-machine replay; and
6. explicit release notes comparing the old and new claims.